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Special Collection:

Science from the Surface Water and Ocean Topography Satellite Mission

Key Points:

- Iterative SWOT-fitted bathymetry cuts global median water surface elevation (WSE) bias from 3.10 to 0.18 m in two runs
- Global median flood-wave timing is preserved (median $r = 0.38$); modest variance increase (standard deviation ratio $1.07 \rightarrow 1.11$)
- Comparisons against Hydroweb and GRDC data confirm WSE bias drop and preserved global-scale flood-wave amplitude and timing

Supporting Information:

Supporting Information may be found in the online version of this article.

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SWOT-Fitted River Bathymetry for Global Hydrodynamic Models

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Abstract We present a robust, global method to reconcile absolute water surface elevations (WSE) from the Surface Water and Ocean Topography (SWOT) mission with a large-scale hydrodynamic model by iteratively correcting river bathymetry. After removing sensor and representativeness bias, we update riverbed elevations under DEM and min-depth feasibility constraints and interpolate corrections along reaches. Applied worldwide at 0.05° resolution over 2023–2025, the approach reduces the global median absolute WSE bias (3.10 m $\rightarrow$ 0.18 m), preserving the same global median correlation (0.38). A modest ~4% increase in the standard deviation ratio (1.07 $\rightarrow$ 1.11) indicates slight variance inflation. Independent evaluation over 2015–2025 at 16,746 Hydroweb sites and 2123 GRDC gauges, restricted to reaches with updated beds, confirms substantially reduced bias, negligible change in both correlation and variance, and no significant change in global median discharge skill. When combined with models, this bias-aware bathymetry will enable a refined numerical representation of floods globally.

Plain Language Summary Since its launch in 2022, NASA's Surface Water and Ocean Topography (SWOT) mission has been providing river water levels at global scale. When we compare satellite measurements to model results, disparities show up because hydrological models often use terrain maps that contain vertical errors. This prevents us from using SWOT data to improve the models. We present a robust method that adjusts the model's river bottom elevations so that modeled water levels align with SWOT. The method removes known reference and location mismatches, then shifts river bottoms and repeats the model run until the differences reach an acceptable minimum. Applied worldwide at 5-km resolution, this approach reduces the typical (median) water-level error from 3.10 to 0.18 m after two cycles, without changing overall flood-wave timing. A small increase in variability is observed. Checks using data from other satellites and thousands of stream gauges confirm the large reduction in bias and little change in streamflow accuracy. The proposed method prepares models to directly ingest SWOT water levels in the future, where we expect further gains in flood timing and magnitude, supporting better flood risk assessment and water management at continental scales.

1. Introduction

Accurate representation of river channel geometry and, in particular, riverbed elevation (bathymetry), is a first-order control on flood dynamics in hydrodynamic models (Bates, 2023; Nguyen et al., 2026). Channel geometry, defined here by width, depth, and bed elevation, governs river conveyance and the partitioning of flow between in-channel and overbank pathways. Underestimated geometry (narrow or shallow channels) promotes premature overflow and overestimation of flood extent, whereas overestimated geometry (wide or deep channels) suppresses overflow and leads to underestimation of inundation. Consequently, errors in riverbed elevation propagate directly into biases in flood magnitude, extent, and timing, even when hydrologic forcing is accurate (Decharme et al., 2012; Yamazaki et al., 2011).

Flood simulations also depend on the accuracy of surface and sub-surface runoff generation, typically derived from land surface models. These fluxes are subject to uncertainties in forcing and model structure, and are often improved through bias correction or data assimilation (Collins et al., 2024; Paiva et al., 2013). Satellite observations of water surface elevation (WSE) provide a direct constraint on river stage and dynamics and, when combined with channel geometry, on river storage. In particular, the Surface Water and Ocean Topography

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(SWOT) mission offers near-global WSE observations at scales relevant for large rivers, creating new opportunities to constrain hydrodynamic models.

However, both flood modeling and the effective use of satellite observations are fundamentally limited by uncertainties in riverbed elevation. While river width and surface extent can be estimated from remote sensing (Feng et al., 2022; Pavelsky & Smith, 2008; Yamazaki et al., 2014), bathymetry remains largely unobservable at large scales (Andreadis et al., 2013; Jiang et al., 2021). Existing approaches to infer riverbed elevation, such as rating-curve inversion, optimization, and data assimilation (Andreadis et al., 2007; Durand et al., 2008; Getirana et al., 2009; Neal et al., 2021; Yoon et al., 2012), are often data-intensive or region-specific, which may limit their scalability. As a result, large-scale hydrodynamic models rely on simplified parameterizations of channel geometry, introducing systematic errors in simulated water levels and storage (Decharme et al., 2012; Getirana et al., 2012; Luo et al., 2017; Yamazaki et al., 2011).

These structural errors have two key consequences. First, they degrade flood simulations by misrepresenting river storage and overflow thresholds, leading to biases in flood extent and timing (Neal et al., 2021). Second, they introduce meter-scale offsets between modeled and observed WSE, limiting the integration of satellite observations (Getirana et al., 2013; Yamazaki et al., 2012). Anomaly-based approaches can remove mean biases but discard absolute elevation information that governs hydraulic gradients, backwater effects, and floodplain connectivity. Addressing riverbed elevation errors is therefore essential for both physically consistent flood modeling and the use of satellite-derived WSE. Improving bathymetry aligns modeled and observed water levels while restoring realistic surface water dynamic properties that control flooding.

Here, we propose a global approach to estimate riverbed elevation by leveraging systematic differences between observed and simulated WSE. The method decomposes these differences into components, and iteratively updates riverbed elevations to minimize bias. By directly targeting structural errors in bathymetry, the approach has the potential to improve flood representation while enabling the consistent use of SWOT WSE observations. We demonstrate the method using the Hydrological Modeling and Analysis Platform (HyMAP; Getirana et al., 2012) and evaluate its impact on simulated water levels and discharge across global river networks.

2. Methods

To improve riverbed elevation estimates using SWOT WSE, we decompose the absolute mismatch between SWOT and the model WSE values (ΔH) into three components: (a) geolocation/representativeness bias arising from mismatches between the SWOT sampling location and the model/DEM location (b^{loc}); (b) systematic sensor biases (b^{sen}); and (c) residual model-parameter bias attributable primarily to riverbed elevation and related geometry (b^{par}). Formally, for reach r and iteration t :

$$\Delta H_{r,t} = \bar{H}_r^{\text{SWOT}} - \bar{H}_{r,t}^{\text{model}} = b_r^{\text{loc}} + b_r^{\text{sen}} + b_{r,t}^{\text{par}} + \varepsilon_{r,t} \quad (1)$$

where H^{SWOT} and H^{model} are the SWOT and modeled WSE, and ε captures random measurement and model errors. The workflow proceeds sequentially. We first estimate and remove b^{loc} and b^{sen} . The remaining difference is attributed primarily to b^{par} , which is used to update riverbed elevation under physical constraints. These updates are then propagated along the river network and the model is rerun iteratively until convergence. Because river hydraulics are nonlinear due to inertia, diffusion, floodplain exchange, backwater, etc., local riverbed edits can induce upstream/downstream changes. As a result, multiple iterations to correct b^{par} are therefore expected. Figure 1 illustrates a hypothetical grid cell and its DEM pixels, as well as the elevation profile of the river pixels and biases assumed in Equation 1.

2.1. Geolocation/Representativeness Bias

River models often define riverbed and WSE at the outlet node of each grid cell (Figure 1a). Depending on the SWOT product used, particularly the Vector Reach Data Product, those outlet nodes may not spatially coincide with satellite data. This is also the case for other radar altimeters (e.g., Jason and Sentinel). In the SWOT reach product, WSE is derived from spatially distributed observations aggregated over a reach segment, rather than corresponding to a single point location, which introduces representativeness differences relative to model outlet

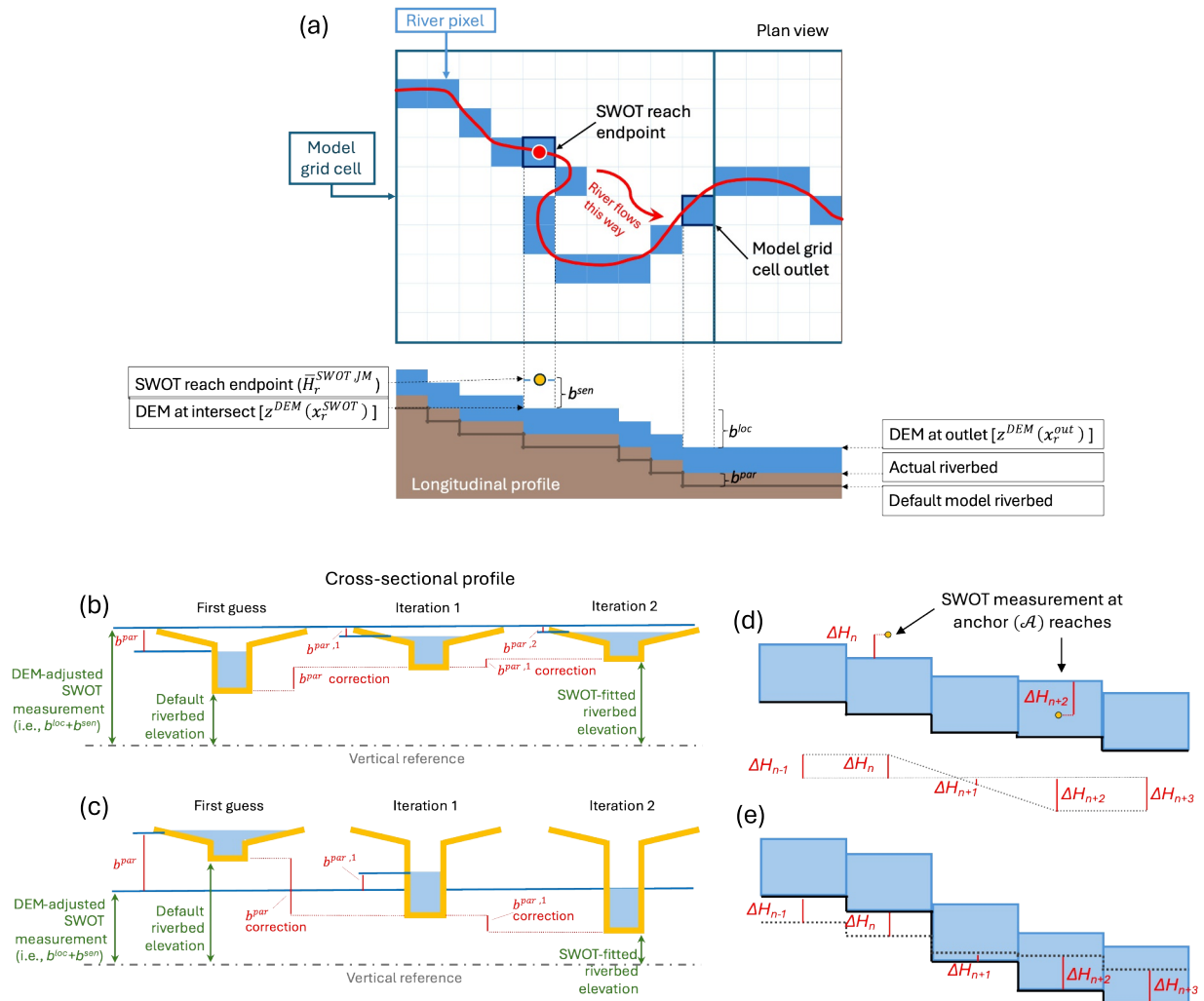


Figure 1. Schematic for the SWOT-fitted riverbed correction: (a) Plan view and longitudinal profile of a hypothetical river within one model grid cell (illustrated with a 10 × 10 DEM sub-grid); WSE and bed are referenced to the cell outlet. (b–c) Iterative bed updates at sample cross-sections where SWOT WSE lies above (b) or below (c) the modeled WSE. (d–e) Along-channel bed profiles before (d) and after (e) linear interpolation of anchor corrections between SWOT-fitted reach endpoints. Abbreviations are defined in Methods.

nodes. Let x_r^{out} be the outlet location of cell x and x_r^{SWOT} the downstream endpoint of a SWOT reach. Using a fine-resolution DEM elevation (z^{DEM}), we quantify the representativeness correction as:

$$b_r^{\text{loc}} = z^{\text{DEM}}(x_r^{\text{SWOT}}) - z^{\text{DEM}}(x_r^{\text{out}}) \quad (2)$$

which we subtract from $\Delta H_{r,t}$ before attributing remaining bias to parameters.

2.2. Systematic Sensor Bias

This term represents a systematic difference between SWOT WSE and model/DEM elevations at the same river location after both data sets have been transformed to the EGM96 vertical datum. The Shuttle Radar Topography Mission (SRTM; Farr et al., 2007) is the DEM basis for many hydrography products (USGS, 2007; Yamazaki et al., 2019), and it was acquired in February 2000. We estimate this cross-sensor bias by averaging SWOT WSE over a seasonally comparable mid-January to mid-March (or JM period) within the mission window. For each reach, we have:

$$\bar{H}_r^{\text{SWOT}, JM} = \text{median}_{JM}(H_r^{\text{SWOT}}) \quad (3)$$

The reach-level systematic offset is then

$$b_r^{\text{sen}} = \bar{H}_r^{\text{SWOT}, JM} - z_r^{\text{DEM}}(x_r^{\text{SWOT}}) \quad (4)$$

which is removed from all SWOT–model comparisons for that reach. This step targets sensor differences and errors associated to SRTM measurements (Simard et al., 2024).

2.3. Residual Parameter Bias and Riverbed-Elevation Updates

After removing b_r^{sen} and b_r^{loc} , we attribute the residual

$$\hat{b}_{r,t}^{\text{par}} = \Delta H_{r,t} - b_r^{\text{sen}} - b_r^{\text{loc}} \quad (5)$$

to bias dominated by model parameterization and update the riverbed elevation as:

$$z_{r,t}^{\text{bed,new}} = z_{r,t}^{\text{bed,old}} + \hat{b}_{r,t}^{\text{par}} \quad (6)$$

We then enforce a physical feasibility constraint to prevent the channel from “vanishing,” that is, a minimum prescribed river depth $d_{\min} = 0.5$ m. In rare cases where updates violate feasibility, an overflow adjustment is applied to maintain a minimum channel depth (Text S1 in Supporting Information S1). In practice, the overflow adjustment is typically ≤ 0.5 m. Note that the constraint assumes rectangular cross-sections and fixed roughness; structural simplifications (e.g., cross-sectional shape, floodplain storage) can also imprint biases and are not explicitly corrected here.

This bias partition assumes that, after representativeness corrections, residuals primarily reflect bathymetry/geometry error. Other structural choices (e.g., rectangular cross-sections, single-value roughness, river width uncertainty, and simplified floodplain storage, in addition to runoff fluxes) may also contribute to bias and are not explicitly corrected here.

2.4. Spatial Interpolation of Riverbed Updates and Iteration

Riverbed updates are first estimated at reaches with reliable SWOT constraints (“anchor” set $\mathcal{A}$, defined as reach endpoints with reliable SWOT observations used to constrain riverbed updates) and are then extended across the network by linear interpolation along flow paths between outlet nodes. Let $c_r \equiv z_r^{\text{bed,new}} - z_r^{\text{bed,old}}$ denote the post-constraint correction at anchor reach $r \in \mathcal{A}$ (i.e., after applying the DEM upper bound and $d_{\min}$; see Section 2.3). Riverbed corrections are interpolated along flow paths between anchor reaches using along-channel distances (see Text S2 in Supporting Information S1).

The updated outlet-referenced riverbed map $\{z_r^{\text{bed,new}}\}$ is used in the next simulation, residuals are recomputed against SWOT, and the cycle repeats until a convergence criterion is met, for example, global-median $|\hat{b}_i^{\text{par}}| < \epsilon$. To avoid overfitting and error compensations, we adopted $\epsilon = 0.2$ m.

2.5. Modeling Framework

HyMAP is a global-scale hydrodynamic model capable of simulating surface water dynamics, including water storage, elevation and discharge in-stream, in rivers and floodplains using the local inertia formulation (Bates et al., 2010). Local inertia represents river flow diffusiveness and the inertia of large water masses of deep flow, which is essential for a physically-based representation of wetlands, lakes, floodplains, tidal effects, and impoundments. HyMAP resolves the local inertia formulation unidimensionally (i.e., a unique flow direction is attributed to each grid cell). Rivers and floodplains interact laterally and have independent flow dynamics, with roughness and geometry derived from land cover characteristics, topography and river parameterization (Getirana, Kumar, et al., 2025).

HyMAP is run globally at 5-km resolution over 2015–2025 using the configuration presented in Getirana et al. (2026) and briefly described below. All elevations are transformed to EGM96 prior to comparison, and only reaches with sufficient SWOT sampling and stable quality flags are used for fitting (see SWOT data processing section below).

Channel geometry and roughness. We prescribe rectangular cross-sections and spatially varying roughness in channels and floodplains globally following Getirana et al. (2012). River width, w [m], is derived from MERIT-Hydro at 3-arcsec spatial resolution (Yamazaki et al., 2019), and gap-filled with a simple discharge-based function (see details in Text S3 in Supporting Information S1).

Topography, hydrography, and forcings. River network and elevations are also derived from MERIT-Hydro processing. HyMAP was forced with daily surface runoff and baseflow from the Global Land Data Assimilation System Version 2.2 (GLDAS-2.2). GLDAS-2.2 runoff was bias-corrected using a gauge-based multiplicative scaling approach inspired by Collins et al. (2024), in which correction factors are derived from observed–simulated discharge mismatches and applied over upstream drainage areas (see details in Text S4 in Supporting Information S1).

Copernicus Marine Service's altimetry-based daily sea surface height data (Taburet et al., 2019) are used as downstream boundary conditions along coastlines (171,218 coastal grid cells globally) to represent the impacts of sea level variability on river dynamics (see details in Text S5 in Supporting Information S1).

2.6. SWOT Data Processing

We adopted SWOT WSE observations from the SWOT_L2_HR_RiverSP_2.0 data product between 1 July 2023 and 10 March 2025 for each reach in the SWOT Prior River Database (SWORD) version 16 about twice every 21 days (Altenau et al., 2021). To facilitate downloading such a large number of observations, we used the Hydrocron tool (Greguska et al., 2025). We filtered the downloaded observations using the bitwise quality flag (“reach_q_b”; SWOT, 2024) and an outlier filter. For the bitwise quality flag, we filtered out all observations with a bit >17 , which removes observations with greater degradation (e.g., geolocation quality). For the outlier filter, we calculated the median and standard deviation across the time series for each reach and removed observations that were outside of the median ± 2 times the standard deviation. These filters retained approximately 28% of the original observations (total of 3,634,164 measurements globally).

HyMAP-SWOT connection. The SWORD-mirror network (Sikder et al., 2024) was used as a “connector” between HyMAP grids and SWORD river reaches. SWORD-mirror is based on SWORD version 16 and includes a subset of 241,852 SWORD reaches. We assumed 0.12° as the maximum distance between each SWORD-mirror reach and HyMAP grids. Within the 0.12° radius, a drainage area match was performed, where the closest HyMAP drainage area was assigned to the SWORD-mirror reach. If drainage area mismatch is greater than $\pm 20\%$, the reach is neglected. The process resulted in 188,517 HyMAP-SWORD matches; a 78% retention.

2.7. Model Evaluation

To assess how SWOT-fitted bathymetry impacts model skill independently of SWOT, we compared HyMAP simulations against two global data sets over 2015–2025: Hydroweb (WSE) at 16,746 altimetric locations and GRDC daily discharge at 2123 gauges. We computed the normalized information contribution (NIC; Kumar et al., 2014) for standard metrics relative to the default-bed simulation, that is, Kling-Gupta Efficiency (KGE), relative error (RE), correlation (r), and standard deviation ratio (rstd). Consistent with our WSE evaluation protocol, for Hydroweb we report NIC of unbiased KGE (ubKGE) and $\Delta|\text{bias}|$ in place of KGE and RE. Positive NIC indicates improvement, while negative $\Delta|\text{bias}|$ denotes reduced absolute bias. Analyses were restricted to locations where SWOT data is available and the riverbed was updated, which substantially reduces the number of evaluation sites relative to the full Hydroweb/GRDC catalogs but isolates the direct effect of bathymetry fitting.

River bathymetry was also evaluated against river cross-section surveys at 3,132 locations, 1,262 in Brazil and 1,870 in the contiguous U.S. (CONUS). This is a subset of 6,629 stations available at river reaches affected by the SWOT-fitting approach. Details on the river cross-section data acquisition and processing are provided in Text S6 in Supporting Information S1.

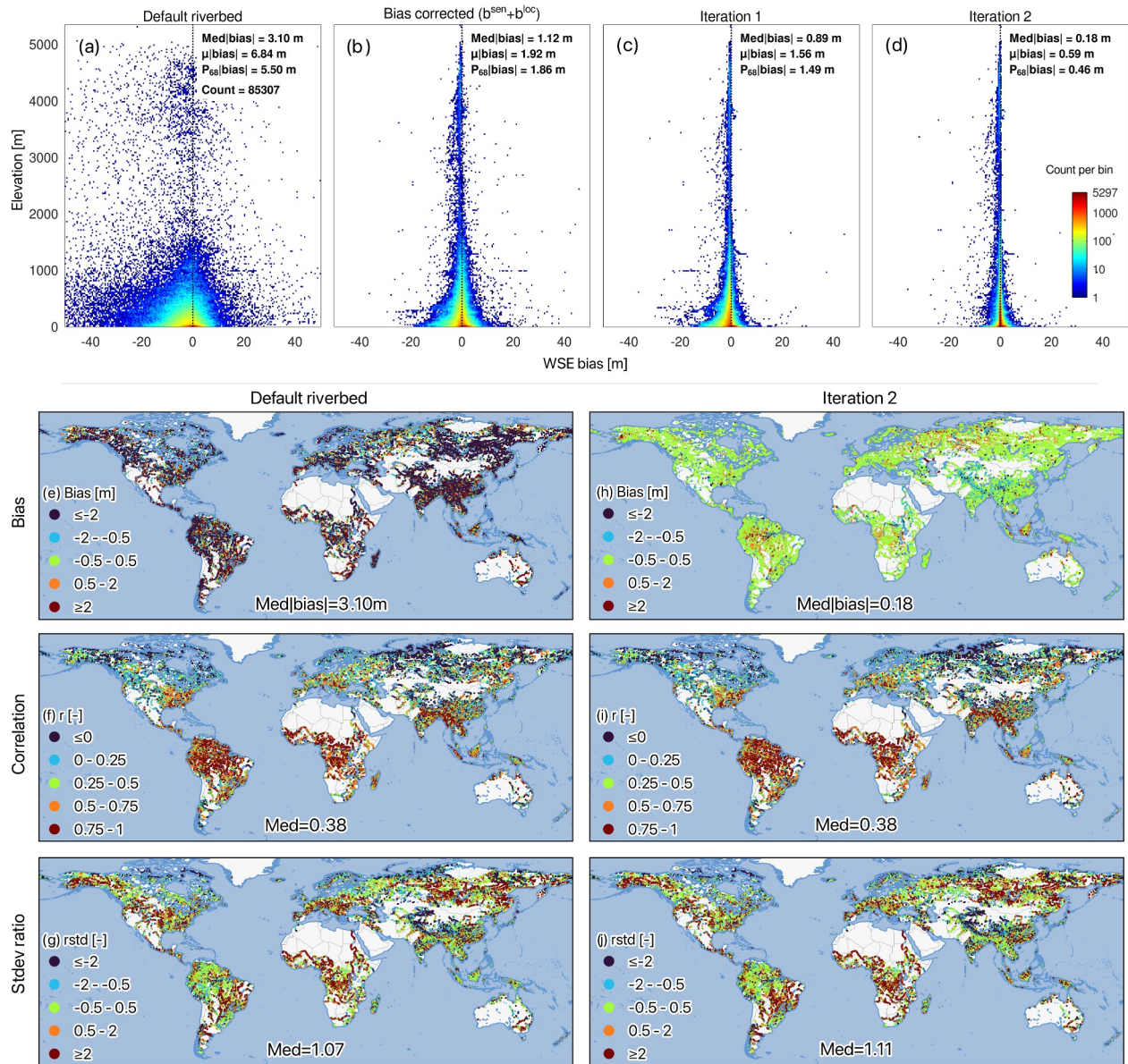


Figure 2. Global SWOT-fitted riverbed. (a–d) Heatmaps of WSE bias between SWOT and HyMAP at 85,307 locations for (a) default riverbed, (b) sensor- and representativeness-corrected WSE (after removing $b^{sen} + b^{loc}$), (c) iteration 1 of bathymetry fitting, and (d) iteration 2. (e–g) Spatial distribution of (e) bias, (f) correlation, and (g) standard deviation ratio for the default riverbed simulation. (h–j) Same three metrics after iteration 2. Here, b^{sen} denotes sensor and b^{loc} denotes geolocation/representativeness bias (see Methods).

3. Results

Comparing HyMAP WSE (default riverbed) with SWOT at 85,307 locations yields a global median $|bias|$ of 3.10 m and a mean of 6.84 m (Figure 2a). Removing b^{sen} and b^{loc} substantially reduces errors (Figure 2b): the median and mean $|bias|$ drop to 1.12 and 1.92 m, respectively. This demonstrates that accounting for representativeness and reference-related offsets substantially reduces the mismatch between SWOT and modeled WSE.

Proceeding with the riverbed fitting, median global biases computed between SWOT and modeled WSE drop to 0.89 m at the first iteration (Figure 2c). After two iterations, the median $|bias|$ falls to 0.18 m (Figure 2d). Large residual biases are concentrated at lower elevations, such as rivers within the central and low Amazon basin, East and West Africa, and Australia. The global median correlation remains unchanged at $r = 0.38$, indicating no overall shift in flood-wave timing; higher correlations cluster in the tropics (e.g., northern South America,

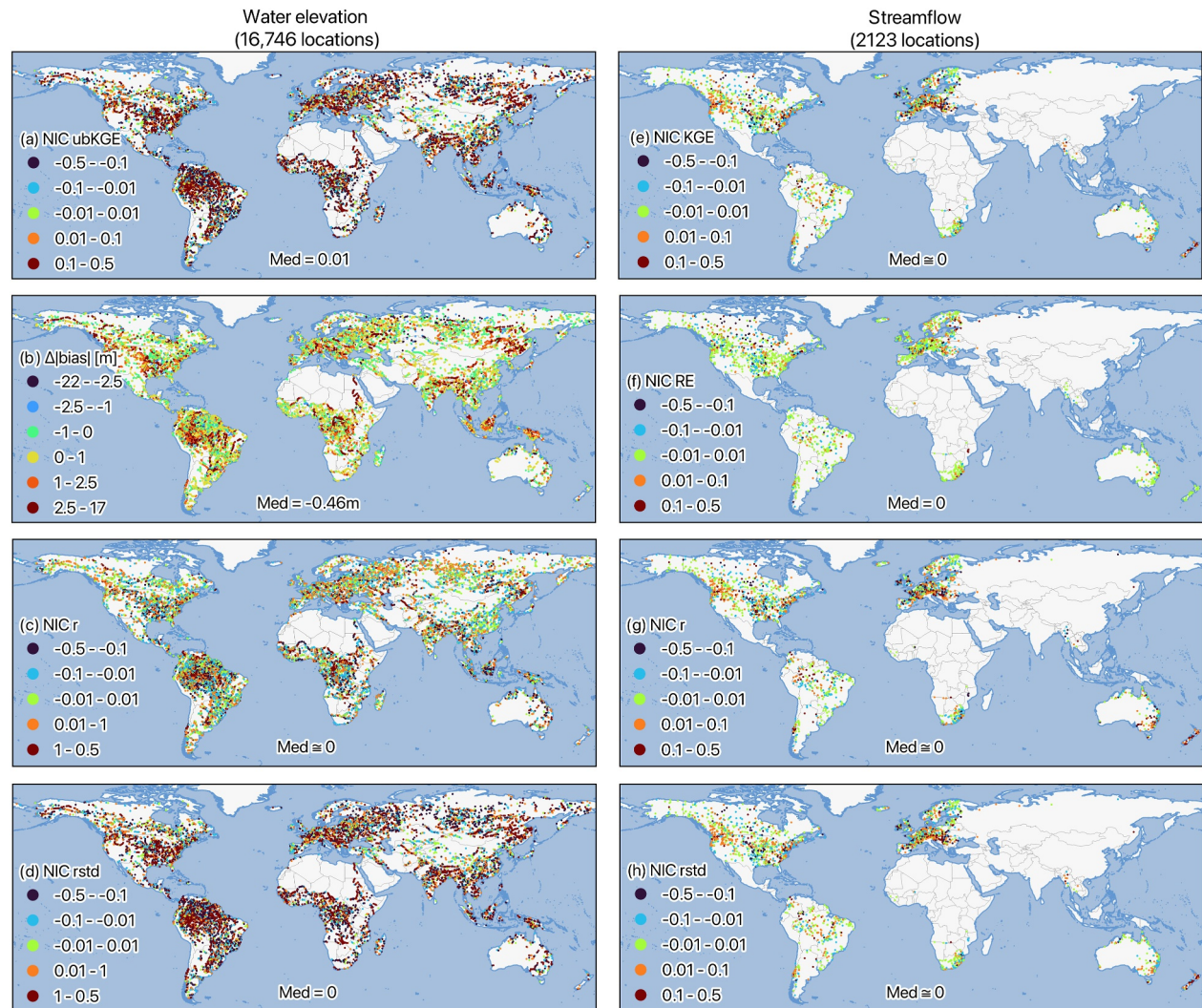


Figure 3. Impact of SWOT-fitted riverbed on simulated WSE and streamflow. Performance changes are summarized using the normalized information contribution (NIC) for key metrics, that is, Kling-Gupta efficiency (KGE), relative error (RE), correlation (r), and standard deviation ratio (rstd), computed relative to the default riverbed. Panels compare simulations against independent data sets: Hydroweb altimetric WSE (left) and GRDC daily discharge (right). For WSE, we report NIC of unbiased KGE (ubKGE) in place of KGE and $\Delta|bias|$ (change in absolute bias) in place of RE.

western/central Africa, south–southeast Asia). In contrast, the standard deviation ratio shows $\sim 4\%$ degradation ($1.07 \rightarrow 1.11$), that is, slightly amplified WSE variability, most evident in western South America, central Africa, and southeast Asia (Figures 2g and 2j).

Mirroring the evaluation against SWOT, comparisons against Hydroweb's water elevation (Figures 3a–3d) indicate a substantial reduction in absolute WSE bias (global median $\Delta|bias| = -0.46$ m). Global median correlation shows no net change ($NIC_r = 0$), confirming that flood-wave timing is not generally degraded by the static bed update. Hydroweb further shows no substantial degradation in variability, with a median $NIC_{rstd} \approx 0$, and correspondingly no change in global ubKGE (median $NIC_{ubKGE} \approx 0$). On the discharge side, no significant change is detected in global medians of KGE, RE, r , or rstd at GRDC gauges, indicating that the bathymetry correction does not materially alter bulk streamflow skill at the global scale.

Regionally, NIC patterns for GRDC discharge reveal coherent, but compensating, signals that explain the near-zero global medians (Figures 3e–3h). NIC_{KGE} shows broad improvements across South America, especially Brazil, along long, low-slope reaches. In contrast, clusters of negative NIC_{KGE} in parts of Europe and North America (U.S. and Canada) indicate localized degradations that offset South American gains. For RE, we find

clear improvements in the Amazon and scattered positive sites in Canada and Europe, but these are counterbalanced by nearby degradations, yielding a global median ≈ 0 . It is important to note that GRDC data distribution is highly heterogeneous across the globe, with large voids in Africa, Middle East, Asia and Russia (Figure S1 in Supporting Information S1 shows examples of streamflow simulations globally). This prevents a comprehensive quantification of actual impacts of bathymetry fitting on simulated streamflow at the global scale.

Correlation changes are similarly heterogeneous: net improvements over Brazil (and pockets in Australia) coexist with degradations across portions of North America and Europe, while South Africa is mixed. The standard deviation ratio mirrors this spatial distribution, that is, improved variance behavior in Brazil, but degraded in parts of North America and Europe. Collectively, these maps suggest that while the bathymetry correction successfully restores absolute stage accuracy globally, its secondary effects on streamflow variability and phasing depend on local hydraulics and configuration (e.g., roughness/width priors, management, floodplain connectivity, DEM alignment, and SWOT data availability). This spatial heterogeneity underscores the need for basin-specific tuning to retain the global bias reduction while minimizing regional penalties in dynamical skill.

Comparison of SWOT-fitted bathymetry against surveyed cross-section observations shows reduced height error in both countries, but with opposite starting biases and unequal gains (Figures 4a and 4b). At CONUS gauges, the default bathymetry initially overestimated mean flow depth by 1.05 m on average (81% of gauges over-predicted); after iteration 2, the mean bias drops to 0.08 m (56% over-predicted), mean absolute error (MAE) falls by 16.6% (1.55 $\rightarrow$ 1.29 m), and the fraction of gauges within their reported observational uncertainty rises from 21.1% to 27.3%. At Brazilian gauges, HyMAP instead underestimates mean river height (mean bias of -3.34 m, 7% over-predicted); after iteration 2, the bias drops to -2.89 m (14% over-predicted), MAE falls by 8.5% (3.76 $\rightarrow$ 3.44 m), and the hit rate (i.e., modeled river-height estimates falling within the observational uncertainty interval) nearly quadruples (2.3% $\rightarrow$ 10%). These results demonstrate that the SWOT-fitting approach proposed in this study not only reduces the bias between simulated and SWOT WSE, but also moves river height estimates in the right direction when evaluated against independent, surveyed observations at large spatial scales.

Across five major rivers (Amazon, Mississippi, Nile, Murray and Niger), outlet-referenced profiles show that SWOT-fitted bathymetry introduces more spatially variable river heights than the relatively flat DEM-derived riverbed elevations. These are hydro-geomorphologically more plausible, though we lack independent data to quantify accuracy (Figure 4). Profiles also reveal reservoir signatures that the current workflow does not explicitly treat. On the Nile, the Aswan Dam ($\sim 1,200$ km from the outlet) produces local artifacts (e.g., a slight upstream bed rise near the dam) because the SWOT data set used excludes reservoir WSE. Similar features appear on the Niger, near the Jebba–Kainji cascade ($\sim 1,000$ km). These cases underscore the need to incorporate managed systems (reservoir bathymetry and operations) in future iterations.

Flow velocity responses are generally modest, aligning with the near-zero global change in discharge skill. For the Amazon, mean velocity changes negligibly (1.04 $\rightarrow$ 1.02 m/s) despite riverbed adjustments. A clearer dynamical effect emerges on the Nile, where fitted depths roughly drop by half from a deep default (~ 19 m), likely increasing overbank flow and decreasing mean velocity from 0.78 to 0.72 m/s by iteration 2. Overall, the profiles illustrate how bathymetry fitting can alter local hydraulics (river height, velocity) while preserving basin-scale flow statistics (except in shallow or strongly managed reaches, where targeted treatment of boundaries and infrastructure is necessary).

4. Discussion and Conclusions

This study demonstrates an effective approach to refine global river bathymetry by reconciling WSE from SWOT with a global hydrodynamic model (HyMAP). We decompose observation–model mismatch into sensor biases, geolocation/representativeness, and residual parameter components, then iteratively update riverbed elevations under a DEM/min-depth feasibility constraint. The outlet-referenced geometry used by HyMAP at 5-km resolution and an along-channel interpolation of “anchor” corrections allow the method to operate consistently across heterogeneous river lengths, sampling densities, and hydroclimates while keeping the updates hydraulically coherent along flow paths.

The SWOT-fitting over 2023–2025 achieves the intended target: the global median |bias| drops from 3.10 m (default bed) to 0.18 m by iteration 2, while timing remains essentially unchanged—global median correlation 0.38 for both the default and iteration 2. The principal tradeoff is a modest but widespread amplitude increase: the

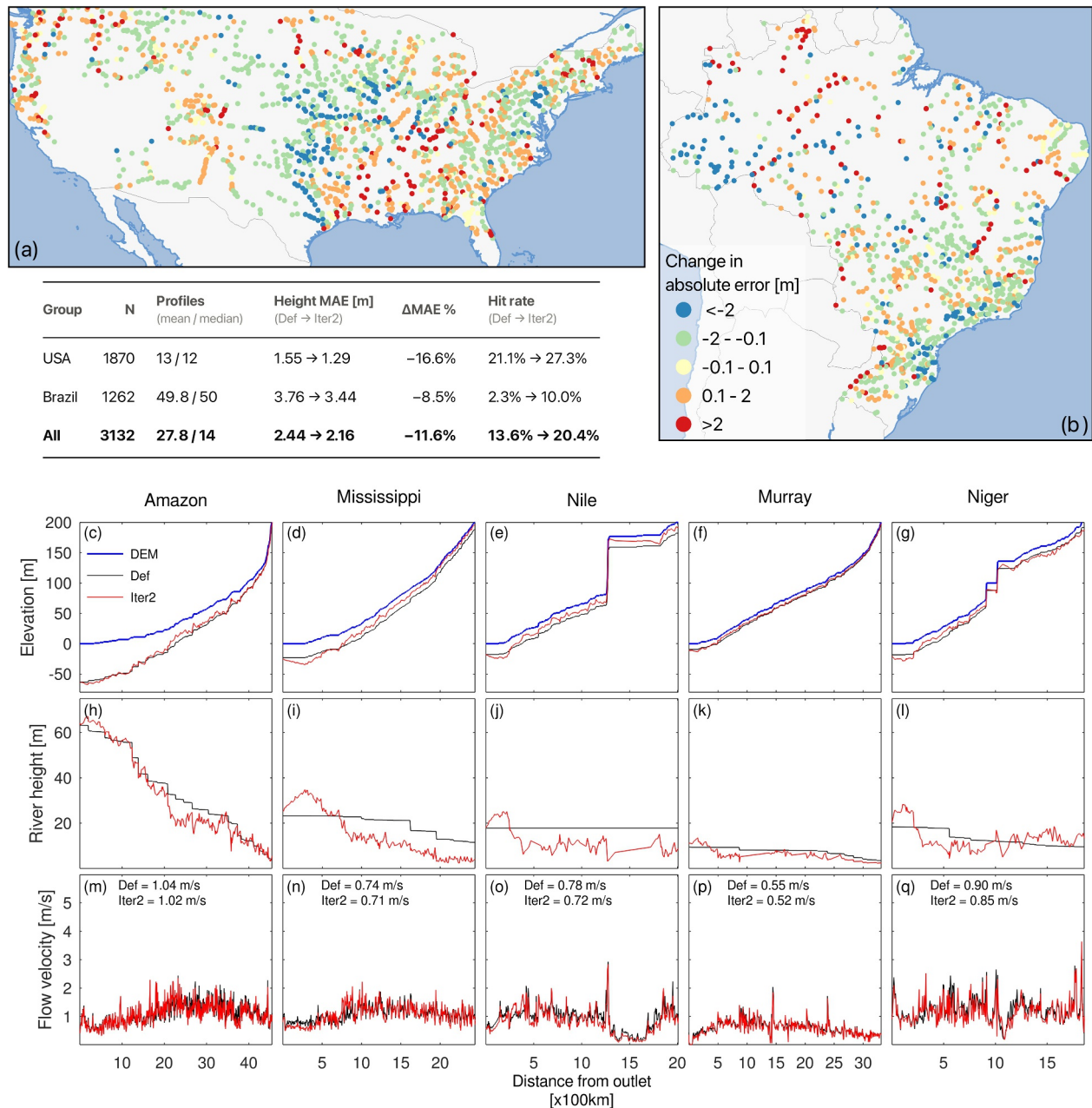


Figure 4. SWOT-fitted riverbed overview. At the top, change in absolute errors between the default (Def) and iteration 2 (Iter2) riverbed parameterizations, when compared to surveys (negative values mean improvement), (a) in the contiguous U.S. and (b) Brazil alongside a table with key metrics, including: number of locations (N), mean and median profile count per location, mean absolute error (MAE) of river height estimates for Def and Iter2, change in MAE (Δ MAE%) and hit rate. At the bottom, longitudinal profiles for selected major rivers: (c–g) riverbed elevation; (h–l) river height; (m–q) flow velocity. Each panel shows results for Def and Iter2. Profiles extend upstream from the river outlet and are truncated at 200 m elevation.

global median standard deviation ratio rises by $\sim 4\%$. Mechanistically, shifting the bathymetry without re-optimizing width/roughness can steepen local WSE gradients or alter backwater extents, increasing variance even as median bias drops.

Independent evaluation over 2015–2025 confirms the SWOT-based findings: substantially reduced bias, unchanged global median correlation and standard deviation ratio, and marginally improved ubKGE, with no significant change in global median discharge skill. In practical terms, the approach restores the absolute level

information (the main constraint for using SWOT magnitudes) with global-scale compensating changes in flood-wave timing and with no significant change in global median discharge skill.

The following factors were adopted to correct riverbed elevation: (a) removing sensor and representativeness components before attributing residuals to bathymetry; (b) enforcing feasibility via a DEM upper bound and a minimum depth of 0.5 m to prevent channels from “vanishing,” with localized DEM lifts only when fitted updates exceed feasible geometry; and (c) performing along-channel interpolation between SWOT-constrained anchors using outlet-to-outlet distances.

Limitations remain. First, sensor bias alignment assumes SWOT WSE averaged over the mid-January to mid-March period is a suitable proxy for stage during SRTM acquisition (February 2000); this seasonal stationarity may be violated where interannual variability, regulation, or morphological change shifts absolute stage. Second, the SWOT record is still short and unevenly sampled, which constrains separation of persistent bias from transient hydrologic states and increases sensitivity to quality control screening. Third, residual misfit reflects broader model uncertainty beyond bathymetry, including meteorological forcing errors, imperfect vertical water and energy fluxes and storages, occasional mismatches between SWORD and HyMAP reaches, and structural simplifications that matter most in (a) tide- and surge-influenced coastal reaches, (b) managed systems with reservoir operations, water withdrawals, and flood control structures, and (c) deltaic bifurcations. Uncertain river widths also propagate into the hydrodynamic modeling. Finally, implementation details are model-specific, so replication across other models and data sets is needed, ideally leading to an ensemble of riverbed elevation fields with quantified uncertainty. Hence, future SWOT-fitted riverbed studies should focus on sensitivity analyses of multiple methods to determine global river bathymetry, including rating curves and traditional optimization and assimilation, while benefitting from sustained SWOT data acquisition. Such studies should also account for sub-daily coastal interactions and human activities, as these processes may lag flood waves and add uncertainty to the bias correction.

Because riverbed elevation controls channel storage, hydraulic gradients, backwater effects, and the transition from in-channel to overbank flow, reducing bathymetric errors is essential for more realistic simulation of flood magnitude, extent, and timing. By using SWOT WSE to constrain riverbed elevation while largely preserving flood-wave timing and discharge skill, the proposed approach strengthens both flood modeling and the physically consistent use of satellite-derived WSE. It also provides a more reliable foundation for future assimilation of SWOT WSE and related variables to further improve flood-wave dynamics (Yoon et al., 2026). The integration of such methods with global hydrological models constrained with multi-satellite data sets has the potential to substantially improve our capability to predict extreme hydrological events (Getirana et al., 2024).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

HyMAP is freely available within NASA's Land Information System (LIS) Framework through <https://github.com/NASA-LIS/LISF>. Global Land Data Assimilation System (GLDAS) Catchment Land Surface Model L4 daily $0.25 \times 0.25^\circ$ GRACE-DA1 V2.2 (GLDAS_CLSM025_DA1_D_EP) data is available from NASA's Goddard Earth Sciences Data and Information Services Center (GES DISC) (Li et al., 2020). Altimetry-based daily sea surface heights (SEALEVEL_GLO_PHY_L4_MY_008_047) can be obtained from the Copernicus Marine Service (European Union-Copernicus Marine Service, 2021). SWOT Level 2 River Single-Pass Vector Reach Data Product, Version 2.0 is available from NASA's Earth Data (SWOT, 2024). GRDC data are available from <https://portal.grdc.bafg.de/applications/public.html?publicuser=PublicUser>. Hydroweb data are available on <https://hydroweb.next.theia-land.fr/>. SWORD-mirror is available from (Sikder et al., 2024). River cross-section data are available from Brazil's Hidroweb data portal <https://www.snirh.gov.br/hidroweb/> and USGS' National Cross-Section Database (NXSDB) (Whaling et al., 2026). Global SWOT-fitted river bathymetry and other parameters needed to run HyMAP and replicate results are available from Getirana, Collins, et al. (2025).

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